

AI Full-Stack Engineer interview preparation

DocMind **AI** Interview Study Guide One-Day Edition

The project, the decisions, the essential concepts and the most likely questions — everything you need, readable in a day.

Built from the DocMind AI repository · github.com/hamza0200/qna-assistant-rag

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1. The one-day plan and your pitch

How to spend the day

Time	Read	Goal
Morning (2.5 h)	Sections 1–3	Say the pitch out loud without notes; draw the architecture from memory; defend 10 decisions
Midday (2 h)	Sections 4–5	Explain RAG, embeddings, vector search, prompt injection, async, SSE in plain words
Afternoon (2 h)	Section 6	Answer the 35 questions aloud; time yourself (≤ 1 minute each)
Late afternoon (1 h)	Section 7	Sketch one system design on paper
Evening (30 min)	Section 8	Cheat sheet; prepare 3 behavioural stories

The full 76-page guide ([docs/Interview-Study-Guide.pdf](#)) is your reference if a topic needs more depth.

Answer shape for every question: one-sentence answer → one level deeper → an example from DocMind or a trade-off.

The 60-second pitch

DocMind AI is a full-stack app where you upload PDFs and ask questions about them. Answers stream in token by token and cite the document and page they came from; you can click a citation to see the exact passage.

It's retrieval-augmented generation. On upload, a FastAPI backend extracts text page by page, splits it into overlapping chunks, embeds each chunk with a local model, and stores the vectors in Postgres with pgvector. On a question, I embed it, find the closest chunks *that belong to this user*, and give only those to Claude with strict rules: answer only from these sources, cite them, say "not found" otherwise. If nothing relevant is found, I don't call the LLM at all.

The frontend is Next.js with TypeScript; the stream is Server-Sent Events read with `fetch`, so I can send the auth header and support a Stop button. It's hardened like production — per-user data isolation with tests, upload validation by magic bytes, rate limits, and defences against a planted prompt-injection attack — with 113 automated tests, CI, and an eval set where retrieval finds the right source for 21 of 21 questions.

The 2-minute extension (if they say "tell me more")

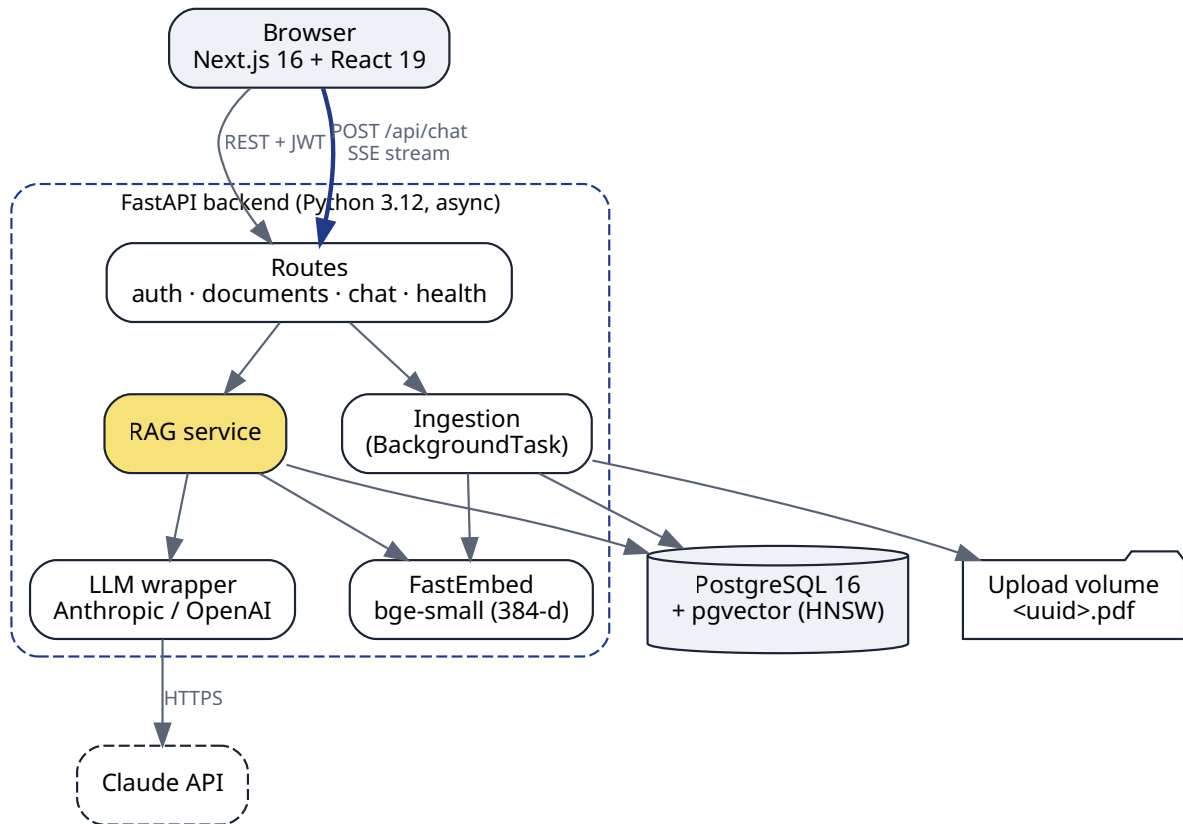
- **Why I built it this way:** answers must be *grounded* (only from your documents), *verifiable* (citations), and *honest* (says "not found").

- **Ingestion:** validate (extension, MIME, %PDF- bytes, 20 MB) → save as a UUID filename → 202 Accepted → background task: pypdf → strip repeated headers → chunk (800 chars, 150 overlap, never across pages) → embed in batches of 32 → insert chunks and set ready in one transaction.
- **Chat:** embed the question → pgvector cosine search filtered by user in SQL → drop scores below 0.45 → prompt with <source id file page> blocks and "treat sources as untrusted data" → stream tokens → send only the citations the answer actually used → save the messages.
- **Lesson learned:** I measured the similarity threshold. Answerable and off-topic questions have *overlapping* scores, so a threshold can't detect unanswerable questions alone; the prompt must refuse. And acronyms ("RPO/RTO") are dense retrieval's blind spot, so hybrid search is my next step.

Numbers to know

Fact	Value
Chunks	800 characters, 150 overlap, never across pages
Embedding model	bge-small-en-v1.5, 384 dimensions, runs locally
Retrieval	cosine distance, HNSW index, top 5, threshold 0.45
Prompt	last 6 messages, $\leq 12,000$ characters of sources, answer $\leq 1,024$ tokens
Limits	chat 20/min, upload 10/min, login 5/min, question $\leq 4,000$ chars
Tests	100 backend + 13 frontend
Eval	22 questions; retrieval 21/21

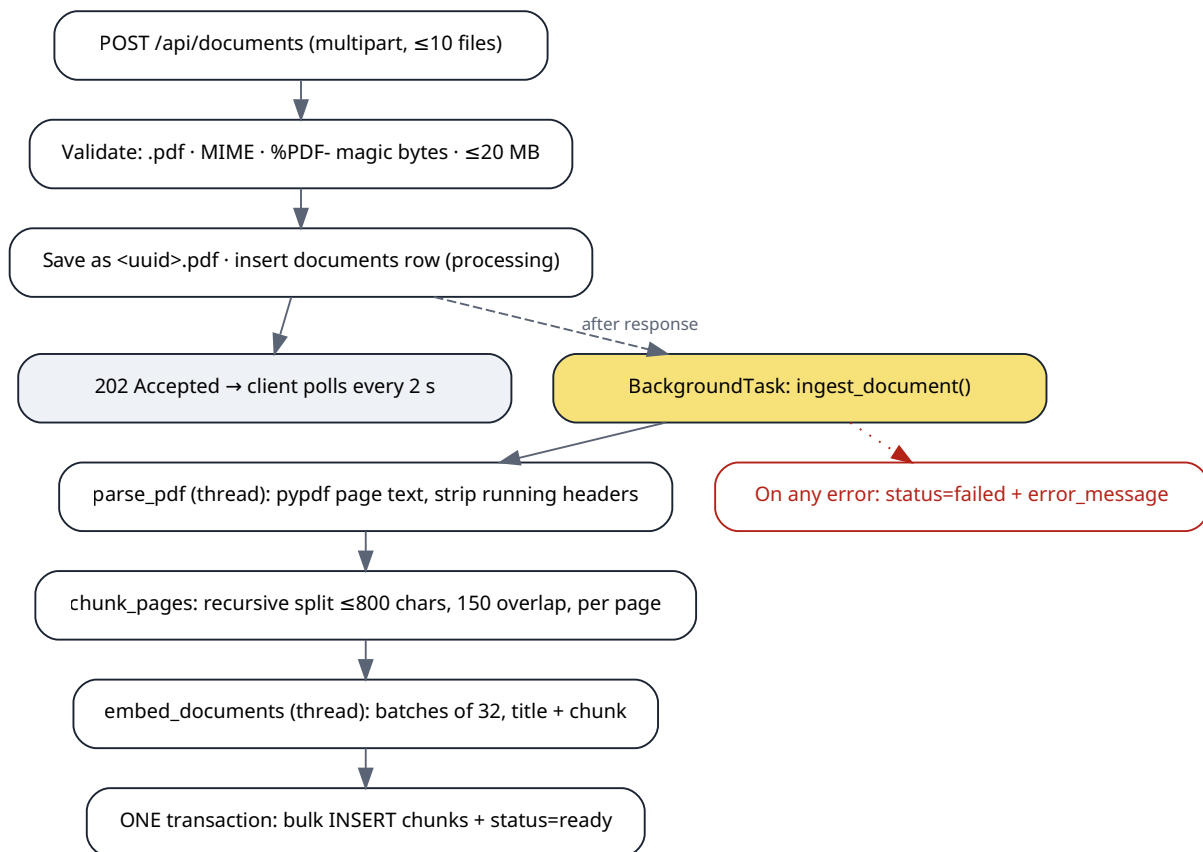
2. How DocMind works



Three tiers: Next.js frontend → FastAPI backend (all the AI logic) → PostgreSQL + pgvector (rows *and* vectors). The only external call is the LLM.

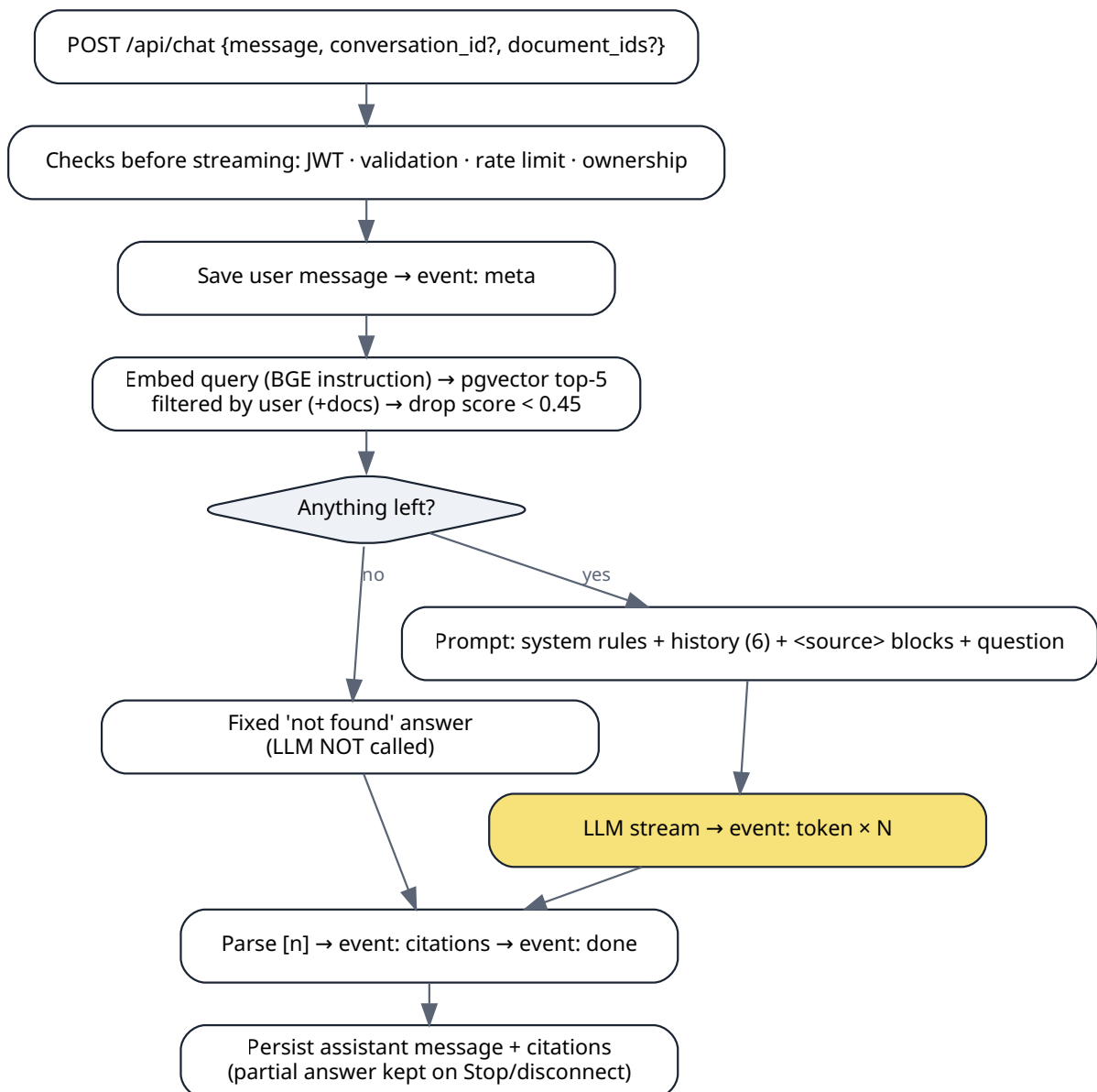
Backend layers: routes (HTTP only) → services (logic) → database. Providers are interfaces (`EmbeddingProvider`, `LLMProvider`), so Claude/OpenAI are swapped by config and tests use fakes.

Upload → ingestion



Step	Where	Key point
Validate	<code>utils/upload_validation.py</code>	extension + MIME + %PDF- magic bytes + size, whole batch first
Store	<code>utils/storage.py</code>	<uuid>.pdf — user filename never used as a path
Respond	<code>routes/documents.py</code>	202 Accepted, status=processing, background task scheduled
Parse	<code>services/pdf_parser.py</code>	pypdf per page, in a thread (CPU-bound); strip repeated headers
Chunk	<code>services/chunker.py</code>	paragraph → line → sentence → word; 800/150; per page
Embed	<code>services/embeddings.py</code>	batches of 32, in a thread; title + chunk is embedded
Save	<code>services/ingestion.py</code>	chunks + ready in one transaction; errors → failed + message

Question → streamed answer



1. **Before streaming:** JWT, validation, rate limit, conversation ownership → normal 4xx errors.
2. Save the user message → `meta` event.
3. **Retrieve** (`services/retriever.py`): `ORDER BY embedding <=> query LIMIT 5`, joined to the user's ready documents, then drop scores < 0.45. Follow-ups also search "previous question + question".
4. **Nothing left?** Send a fixed "not found" answer; **the LLM is not called**.
5. **Prompt** (`services/prompts.py`): system rules + history + numbered `<source>` blocks + question last.
6. **Stream** (`services/llm.py`): each text delta becomes a `token` event.
7. **Citations:** find `[n]` in the answer → send only those → `done`.
8. **Save** the answer. On Stop/disconnect, save the partial answer (in a shielded cancel scope).

SSE events: `meta` → `token × N` → `citations` → `done` (or `error` with a code such as `LLM_TIMEOUT`).

Data model (5 tables)

`users` → `documents` → `chunks` (`embedding vector(384)`, `HNSW index`, `ON DELETE CASCADE`);
`users` → `conversations` → `messages` (`citations` as JSONB). Every query joins back to `user_id` — that's the security model.

3. The decisions that matter

For each: **why this**, the **alternative**, and **when I'd switch**. Say all three and you sound senior.

Decision	Why this	Alternative (cost)	When I'd switch
FastAPI (Python)	Best AI ecosystem; async; validation + docs from type hints	NestJS (one language, but AI libs are Python-first)	TypeScript-only team, AI is a thin API call
pgvector in Postgres	One datastore: transactions, user filter as a SQL join, one backup	Pinecone/Qdrant (sync between two stores, more infra)	100M+ vectors or vector load must scale separately
Local embeddings (FastEmbed)	Free, private, no key, fast, small vectors	OpenAI/Voyage embeddings (cost, data leaves server)	Multilingual or eval shows retrieval is the bottleneck
RAG, not fine-tuning	Fresh, per-user, citable facts; no retraining	Fine-tuning (bad at facts, can't cite, mixes users' data)	Need a fixed style/format at high volume — combine both
Chunks 800/150, per page	Focused embeddings, boundary overlap, exact page citations	Fixed windows (cut sentences), semantic chunking (costly)	Long manuals → larger chunks + parent retrieval
Threshold 0.45 (measured)	Keeps all answerable questions (lowest 0.508), filters clearly off-topic	0.35 filters nothing; 0.52 drops a real question	A reranker gives calibrated scores to threshold on
SSE via fetch	One-way streaming over plain HTTP; auth header; Stop via abort	WebSockets (stateful, overkill); EventSource (GET only, no headers)	Real-time two-way features
BackgroundTasks	No extra infra; 202 + polling	Celery/SQS workers (durable, retries, more ops)	Multiple servers, big files, need retries
Provider-agnostic LLM wrapper	Swap Claude/OpenAI by config; one error model	LangChain (more abstraction than needed)	Many integrations / agent graphs
Cite only referenced sources	A chip means "this answer used this passage"	Show all retrieved (implies false support)	Single provider with native citations
JWT in localStorage	Simple for an SPA on a separate API origin	httpOnly cookie (no JS access, but needs CSRF protection)	Production: cookie + refresh tokens
Rate limits in memory	One decorator per route; per user, not per IP	Redis / API gateway	More than one backend replica

The honest summary: every simplification (background tasks, localStorage token, in-memory limits, local files) has a documented production replacement (queue, cookie, Redis, S3) in `docs/DECISIONS.md`.

4. AI essentials

Tokens. Models read sub-word pieces (~4 characters each in English). You pay per token (output costs more than input), limits are in tokens, and latency grows with output length.

Context window. The total tokens the model can see (prompt + answer). Bigger isn't free: cost, latency, and worse use of information in the middle of long prompts. DocMind sends ~5 chunks, 6 history messages, $\leq 1,024$ answer tokens.

Temperature. Low = focused and deterministic, high = varied and error-prone. Use low for factual Q&A. Newer models may replace it with an "effort" setting (DocMind: `effort: low`).

Embeddings. Text $\rightarrow$ vector; similar meaning $\rightarrow$ close vectors. **Cosine similarity** = angle; on normalized vectors it ranks the same as dot product and L2. BGE models want an instruction prefix on *queries* only.

RAG in five steps. Ingest (parse, chunk, embed, store) $\rightarrow$ retrieve (embed question, nearest neighbours + filters) $\rightarrow$ augment (prompt with numbered sources) $\rightarrow$ generate (stream, cite) $\rightarrow$ post-process (citations, save, log).

Chunking. Too big = blurry embeddings and wasted tokens; too small = no context. Recursive splitting on natural boundaries + overlap is the solid default.

Hybrid search + RRF. Keyword search finds exact terms (acronyms, codes); vectors find meaning. Merge the two ranked lists with Reciprocal Rank Fusion: $\text{score} = \Sigma 1 / (60 + \text{rank})$.

Reranking. A cross-encoder reads question and chunk *together* — far more accurate than vector similarity, too slow for everything. Retrieve 20–50, rerank to 5.

Query rewriting. Turn "and what about Enterprise?" into a standalone question before retrieval (or DocMind's cheap version: also search with the previous question).

Hallucination. Fluent content not supported by sources. Reduce it: skip the LLM when nothing is retrieved, "answer only from sources / say what's missing", require citations, low randomness, evaluate refusals. You reduce and expose hallucinations; you can't eliminate them.

Prompt engineering. Clear role, explicit rules, delimiters separating instructions from data, question last, a defined output format, edge cases handled (missing, partial, conflicting info), and a version number — prompts are code.

Tool calling. The model asks your code to run a declared function with JSON arguments; *your code* validates, authorizes and runs it. **Agents** are an LLM in a loop of tool calls — use only when steps can't be fixed in advance; add limits and human approval for risky actions.

Structured outputs. Enforce a JSON Schema via the API, or validate with Pydantic/Zod and retry.

Fine-tuning vs RAG vs prompting. Prompting = behaviour; RAG = knowledge; fine-tuning = style/format/narrow tasks at scale.

Evaluation. Measure retrieval (hit-rate@k) and generation (correct facts, faithfulness, citations, correct refusals, injection resistance) separately. LLM-as-judge scales but is biased — validate it against human labels.

Vector indexes. **HNSW:** layered graph, great recall, handles inserts, more memory (DocMind's choice). **IVFFlat:** clusters, faster to build, needs data first, lower recall. Both are *approximate* — you trade a little accuracy for a lot of speed.

5. Engineering essentials

Backend

- **Event loop:** one thread juggles many waiting tasks. **Never block it** — CPU work or blocking calls go to `asyncio.to_thread` (DocMind: PDF parsing, embeddings) or to worker processes.
- **Dependency injection:** `Depends(get_current_user)` / `Depends(get_db)` ; tests swap them with `dependency_overrides`.
- **Status codes:** 201 created, **202 accepted (async work)**, 204 deleted, 401 not logged in, 403 not allowed, **404 for other users' data** (don't reveal it exists), 409 conflict, 413 too large, 415 wrong type, 422 invalid, 429 rate limited.
- **SSE vs WebSockets:** SSE = one-way over plain HTTP (streaming answers); WebSockets = two-way and stateful (chat apps, collaboration).
- **Idempotency:** repeating a request has the same effect; use an `Idempotency-Key` for POSTs such as payments.
- **Pagination:** offset (simple, slow on deep pages) vs cursor/keyset (fast, stable).

Frontend

- **Server vs client components:** server = rendered on the server, no JS shipped; client (`"use client"`) = state, effects, events. DocMind's app pages are client components because the token lives in the browser.
- **SSG / SSR / CSR:** build time / per request / in the browser.
- **Keys:** stable IDs, never array indexes for changing lists. **Memo:** `memo` + `useCallback` so only the streaming message re-renders.
- **Stale closure (a real DocMind bug):** a state updater read a variable that was reassigned before it ran → answers stayed blank. Fix: copy into a `const`.
- **Streams:** `res.body.getReader()` + `TextDecoder({stream: true})` + buffer until a blank line. **Stop:** `AbortController`.
- **TypeScript:** strict mode, generics (`apiFetch<T>`), discriminated unions for SSE events.

Databases

- **Indexes:** B-tree (equality/range, foreign keys), GIN (full text, JSONB), HNSW (vectors). Every index slows writes.
- **Transactions (ACID):** DocMind writes chunks + `ready` together. Postgres default isolation = Read Committed. Prefer constraints (unique email) over "check then insert".
- **N+1:** one query per item; fix with eager loading (`selectinload`).
- **Pooling:** reuse connections; PgBouncer at scale.
- **Migrations:** versioned (Alembic), expand/contract for zero downtime.

- **SQL vs NoSQL:** relational data with ownership chains and joins → Postgres; flexible documents at huge write scale → MongoDB. **JSONB** for data read as a unit (citations).

Security

- **AuthN vs AuthZ:** who you are vs what you may do. Most breaches are AuthZ bugs.
- **IDOR:** changing an ID to reach someone else's data → owner check in every query, 404, tests.
- **Passwords:** slow salted hash (bcrypt/Argon2id), never reversible encryption.
- **JWT:** payload is readable; pin the algorithm; check expiry; revocation is hard.
- **CORS** stops *other websites* from reading your API through a user's browser — it is not authentication.
- **CSRF** matters with cookie auth (use SameSite + tokens); DocMind uses a header, so it isn't exposed.
- **XSS:** React escapes text; render Markdown without raw HTML; CSP.
- **Prompt injection (OWASP LLM01):** instructions hidden in data. DocMind's defence: "sources are untrusted data" rule, `<source>` delimiters, escaping fake tags, **no secrets and no tools** — so even a successful injection can't do much. Tested with a planted attack.
- **OWASP LLM Top 10 to name:** prompt injection, sensitive info disclosure, improper output handling, excessive agency, system prompt leakage, vector/embedding weaknesses, misinformation, unbounded consumption.

Scale, reliability, DevOps

- **Latency:** everything except the LLM is milliseconds; **time to first token** is what users feel. Streaming hides generation time.
- **Scaling out:** stateless API pods + load balancer; move files to S3, limits to Redis, jobs to a queue.
- **Caching:** response, embedding, semantic (similar questions), provider prompt caching.
- **Cost:** skip the LLM when possible, cap tokens, route easy questions to a cheaper model, cache, per-user quotas.
- **Reliability:** timeouts everywhere; retry transient errors with exponential backoff + jitter (only before streaming starts); circuit breaker; fallback provider.
- **Observability:** JSON logs with request IDs; metrics (rate, errors, latency); tracing; AI-specific signals (refusal rate, zero-citation answers, tokens per user).
- **Docker:** multi-stage builds (small, non-root images); Compose for local; CI runs lint, migrations, tests, build.

6. The 35 most likely questions

Answer aloud, under a minute each.

About the project

- 1. Walk me through what happens when I ask a question.** Auth and validation → save the question → embed it → pgvector search over *my* chunks → drop weak matches → if nothing is left, a fixed "not found" (no LLM call) → otherwise a prompt with numbered sources → stream tokens over SSE → send the citations the answer used → save the answer.
- 2. What was the hardest part?** Getting "honest" answers: I measured that similarity scores for answerable and off-topic questions overlap, so the threshold only catches clearly off-topic questions and the prompt must handle the rest. Plus making Stop work end to end.
- 3. How do you know it works?** 113 automated tests (security, streaming, refusals, injection) and an eval of 22 labelled questions — retrieval finds the right source for 21 of 21.
- 4. What would you do next?** Hybrid search, a reranker, follow-up question rewriting, a durable ingestion queue, cookie-based auth, OCR.
- 5. What would you change for production?** Queue + workers, S3, Redis for rate limits, httpOnly cookie auth, tracing and dashboards, a secrets manager, cross-provider fallback.

AI / LLM

- 6. What is RAG, and why not fine-tune?** Retrieve relevant text and answer from it: fresh, private, citable. Fine-tuning teaches style, not reliable facts, can't cite, and would mix users' data.
- 7. What's an embedding? Why cosine?** Meaning as a vector. Cosine compares direction; on normalized vectors it ranks the same as the dot product.
- 8. How did you choose chunk size?** ~800 characters (~200 tokens) + 150 overlap, split on natural boundaries, never across pages. Focused, enough context, exact page citations.
- 9. How do you stop hallucinations?** Don't call the LLM without context; "only from sources, say what's missing"; mandatory citations shown to the user; low randomness; refusal questions in the eval.
- 10. How did you pick the similarity threshold?** Measured: answerable 0.508–0.84, off-topic 0.416–0.541. 0.45 keeps every real question.
- 11. What's hybrid search?** Keyword (exact terms) + vector (meaning), merged by rank with RRF. It fixes acronyms like "RPO/RTO", my weakest eval question.
- 12. What does a reranker do?** A cross-encoder scores (question, chunk) pairs together — more accurate; run it on the top 20–50 candidates.

13. How do you evaluate a RAG app? Retrieval (hit-rate@k) and answers (facts, faithfulness, citations, refusals, injection) separately, on a versioned set, after every change. LLM-as-judge for paraphrases, validated against humans.

14. What is prompt injection, and how did you defend? Instructions hidden in data. Untrusted-data rule, `<source>` delimiters, escaped fake tags, and least privilege: no secrets in the prompt, no tools, per-user data, output never executed. Tested with a planted attack.

15. Tool calling and agents? The model requests a function call; my code validates, authorizes and runs it. Agents loop over this — only when steps can't be fixed in advance, with limits and approvals.

16. How do you control LLM cost? Skip calls when possible, cap context and output tokens, cache, route to cheaper models, per-user quotas, and log tokens per request.

Backend

17. Why FastAPI? Async, validation and docs from type hints, and Python's AI ecosystem.

18. What if you call blocking code in `async def`? It freezes every request on that worker. Use `to_thread` or worker processes.

19. Why 202 for uploads? Accepted; processing continues in the background; the client polls for `ready / failed`.

20. SSE or WebSockets? SSE: one-way over plain HTTP, ideal for token streaming. WebSockets: two-way, stateful.

21. BackgroundTasks vs a queue? In-process and simple vs durable, retried and independently scaled. I'd switch at scale.

22. What happens when the user presses Stop? The fetch aborts → the server cancels the stream → the provider stream closes (no more tokens billed) → the partial answer is saved in a shielded scope.

Frontend

23. Server vs client components? Server: no JS shipped, fetches on the server. Client: interactivity. My pages are client components because the token is in the browser.

24. How do you read a stream in the browser? `getReader() + TextDecoder({stream: true})` + buffer events until a blank line; `AbortController` to cancel.

25. How do you avoid re-renders while streaming? Memoize message components, stable callbacks, keep state local.

26. Tell me about a bug you found. A stale closure: a state updater read a variable reassigned before React ran it, so answers stayed blank. An end-to-end test caught it; fix: capture into a `const`.

Databases

27. Why pgvector and not Pinecone? One datastore, transactions, user filters as joins, no syncing. Switch at hundreds of millions of vectors.

28. HNSW vs IVFFlat? Graph vs clusters. HNSW has better recall and handles inserts; IVFFlat builds faster but needs data first.

29. What's N+1? One query per list item; fix with eager loading.

30. Isolation levels? Read Committed (default), Repeatable Read, Serializable. Many races are better solved with a unique constraint.

Security

31. What's IDOR, and how did you prevent it? Changing an ID to reach others' data → owner check inside every query, 404, UUIDs, tests for each resource.

32. How do you store passwords? bcrypt (cost 12), salted, slow; Argon2id is the modern first choice.

33. JWT pitfalls? Readable payload, pin the algorithm (`alg: none` attack), check expiry, revocation is hard, storage trade-offs.

34. CORS, CSRF, XSS in one breath? CORS = other sites can't read my API via your browser; CSRF = forged cookie requests (not an issue with header tokens); XSS = injected script (React escaping, no raw HTML, CSP).

Tricky follow-ups

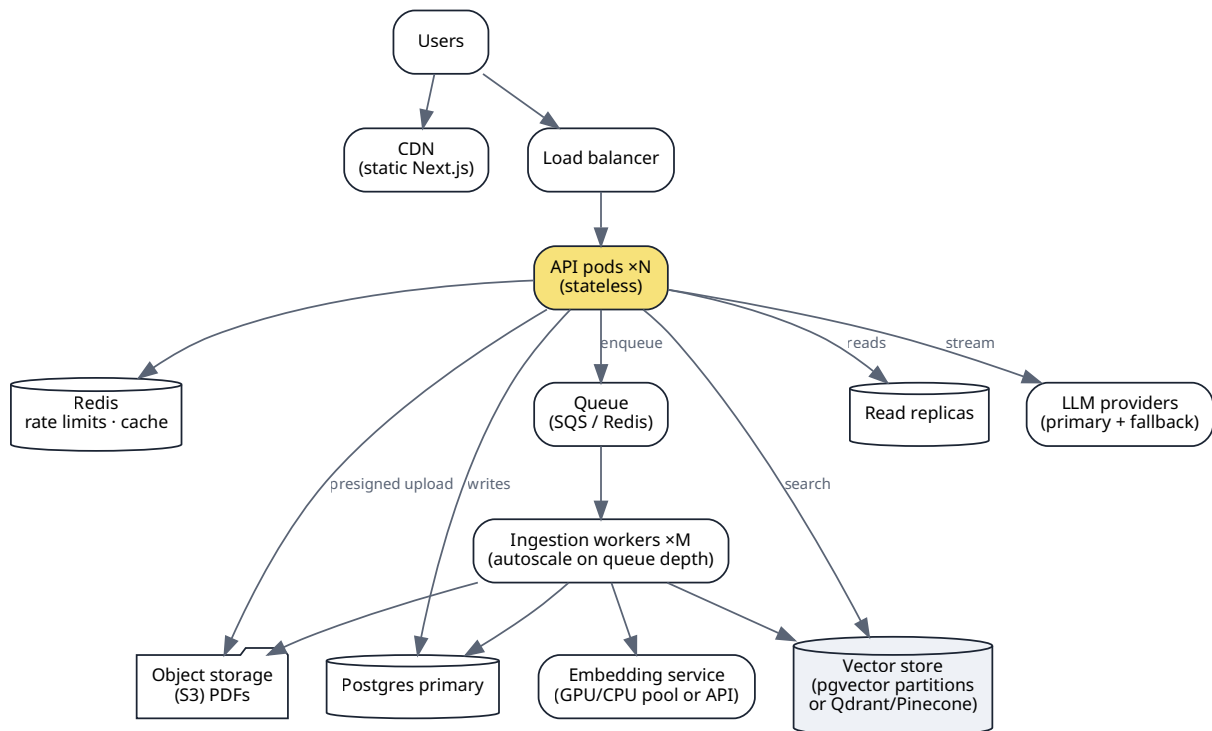
35. Quick answers: - *Scanned PDF?* No text layer → marked `failed` today; add OCR (Tesseract or a cloud OCR service) with page numbers. - *500-page document?* Queue worker, page batches with progress, larger embedding batches, section-aware chunking; retrieval is still fine (~1,500 chunks). - *Multi-language?* Multilingual embedding model + re-embed, answer in the user's language, language-aware keyword search, non-English eval questions. - *PII?* Minimize, per-user access, never log contents, encrypt, check provider data retention, deletion that cascades, optional PII redaction before the LLM. - *Conflicting sources?* The prompt says who says what and prefers authoritative documents; later, rank by date/authority metadata.

7. System design in one hour

The framework (use it for any design question)

1. **Requirements:** what it does + scale, latency, availability, security, cost. Ask questions.
2. **Estimates:** users, requests/second, data size.
3. **API + data model:** the few key endpoints and tables.
4. **Boxes and arrows:** walk one request through.
5. **Deep dives:** the 2–3 hardest parts.
6. **Bottlenecks, failures, trade-offs, monitoring.**

Practice: scale DocMind to 100k users and 10M documents



Estimates. 10M docs × 20 pages × 3 chunks ≈ **600M chunks** → 384 × 4 bytes each ≈ **~920 GB of vectors** (half that with 16-bit `halfvec`). Too big for one Postgres node. Chat load is modest (~7 questions/s at peak); LLM cost and time to first token matter more.

Design.

- CDN for the frontend; **stateless API pods** behind a load balancer.
- **Uploads straight to S3** (presigned URLs) → **queue** → **autoscaled ingestion workers** → **embedding service** (GPU batches or a hosted API).
- **Vectors:** pgvector **partitioned by tenant** (with 16-bit vectors) *or* a dedicated vector DB (Qdrant/Pinecone) kept in sync via an outbox table.
- Postgres primary + **read replicas**; PgBouncer; **Redis** for rate limits and caches.

- **Better retrieval:** hybrid search + reranker.
- **LLM layer:** circuit breaker, fallback provider, prompt caching, model routing, per-tenant token quotas.
- **Observability:** traces, time to first token, refusal rate, tokens per tenant, queue depth.

Bottlenecks → fixes. Vector memory → partitioning/quantization/vector DB. Bulk imports → GPU workers, per-tenant fairness. Filtered search recall for small tenants → partition or exact search. Provider limits → quotas and fallback. Deletion at scale → delete across S3, vectors and rows, with an audit trail.

Two more to sketch on paper

AI support chatbot with tools. RAG over help articles + narrow allow-listed tools (`get_order`, `track_shipment`, `create_return`, `issue_refund`). Validate arguments, **authorize against the logged-in customer** (never trust the model), confirm or hand to a human for risky actions, limit steps per turn, audit log, human handoff with a summary.

Multi-tenant SaaS. Pool (shared tables + `tenant_id` + **Postgres Row-Level Security**) by default; schema-per-tenant or **database-per-tenant** for enterprise customers. Tenant-scoped vector filters and cache keys, per-tenant keys (KMS), quotas and cross-tenant tests.

8. One-page cheat sheet

Pitch: Upload PDFs, ask questions, get streamed answers that cite document + page, and an honest "not found" when the documents don't say. RAG with FastAPI + pgvector + local embeddings + Claude; Next.js frontend; hardened and tested.

Pipeline	Numbers
Parse	pypdf page by page; strip running headers; scanned → failed
Chunk	800 chars, 150 overlap, recursive (¶ → line → sentence → word), never across pages
Embed	bge-small-en-v1.5, 384-d, normalized, batch 32, local ONNX; query prefix instruction
Store	Postgres 16 + pgvector, HNSW vector_cosine_ops, one transaction with status=ready
Retrieve	<=> cosine distance, user filter in SQL, TOP_K 5, MIN_SIMILARITY 0.45
Prompt	system rules + last 6 messages + <source id file page> blocks (≤ 12k chars) + question last
Generate	claude-opus-5-5, effort low, max_tokens 1024, 60 s timeout, 2 retries before stream, refusal fallback
Stream	SSE: meta → token... → citations → done / error; fetch + ReadableStream + AbortController
Cite	parse [n], show only cited sources, chip opens /api/chunks/{id}

Limits: chat 20/min/user · upload 10/min/user; 20 MB, 10 files · login 5/min/IP · question ≤ 4,000 chars · JWT 60 min · bcrypt cost 12.

Quality: 113 tests (100 backend, 13 frontend) · eval 22 questions · retrieval hit-rate 21/21 · threshold measured: answerable 0.508–0.84 vs off-topic 0.416–0.541 (overlap).

Key terms (one line each):

- **RAG** — retrieve relevant text, then generate from it. **Grounding** — answer only from provided sources.
- **Embedding** — meaning as a vector. **Cosine** — angle-based similarity; = dot product on unit vectors.
- **HNSW** — layered graph ANN index; m, ef_construction, ef_search. **IVFFlat** — cluster buckets; lists, probes.
- **Hybrid + RRF** — keyword + vector; merge by $1/(60 + \text{rank})$. **Reranker** — cross-encoder on top-k.
- **Prompt injection** — instructions hidden in data; defend with delimiters, untrusted-data rule, least privilege.
- **IDOR** — access by changing an ID; fix with ownership in every query, 404, tests.
- **SSE** — one-way HTTP stream; **WebSocket** — two-way stateful connection.
- **202 Accepted** — async processing; **idempotency key** — safe retries of POST.
- **Event loop** — never block it; `asyncio.to_thread` for CPU/blocking work.
- **N+1** — fix with eager loading. **ACID**; Postgres default isolation = Read Committed.
- **TTFT** — time to first token, the latency that users feel with streaming.

Talking points (have a story for each):

1. Measured the threshold instead of guessing — and found that a threshold can't detect unanswerable-but-on-topic questions.
2. Prompt injection is contained by design: no secrets, no tools, per-user data, output never executed — plus tests and an eval question.
3. Stop button end to end: AbortController → server cancellation → provider stream closed → partial answer saved in a shielded scope.
4. Stale-closure bug caught by an end-to-end test, not unit tests.
5. Every simplification has a documented production path: BackgroundTasks → queue, localStorage → httpOnly cookie, in-memory limits → Redis, local files → S3.

Next steps: answer eval with live model → hybrid search → reranker → query rewriting → durable queue → cookie auth → OCR → feedback + semantic cache.

Your questions for them: How do you evaluate AI features? Path from prototype to production? Biggest technical challenge this year?